

Semântica na Web

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Contexto ...

Carlos Bazilio
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[Mount Eliza Secondary College](#)

www.mesc.vic.edu.au/ ▾

A Victorian Government school. Includes news, events, curriculum information, college history, information for parents, International Students, virtual tour, staff, ...

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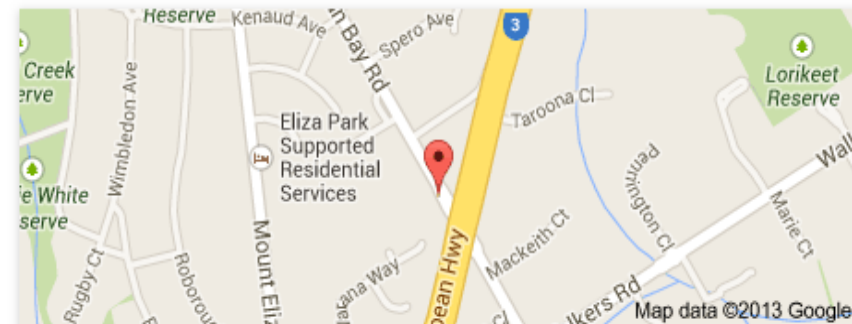
<https://www.michigan.gov/uia/0,4680,7-118-1328-137908--,00.html> ▾

In 1937 we started out as the Michigan Unemployment Compensation Commission. Later, we became the **Michigan Employment Security Commission** or **MESC** ...

[MESC 2013 | MESConference](#)

www.mesconference.org/mesc-2013/ ▾

Mark yourcalendar for the 2013 **Medicaid Enterprise Systems Conference** ... and exhibiting will be posted to the **MESC** website as they become available.



Mount Eliza Secondary College

[Directions](#)

Mount Eliza Secondary College is a public co-educational secondary school located in Mount Eliza, Victoria, Australia. As of 2011 the school caters to approximately 1,000 students in years 7 to 12 from the local area. [Wikipedia](#)

Address: Canadian Bay Rd, Mt Eliza VIC 3930, Australia

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<https://www.facebook.com/mestrado.mescuff> ▾

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M.E.S.C. - Master in Materials for Energy Storage and Conversion, PDF English · http://www.u-picardie.fr/mundus_MESC. MA LLL - European Masters in ...

[\[PDF\] Edital - Mestrado - Revisado em reuniao de 11-09-2012 ...](#)

www.puro.uff.br/sites/default/files/EditalMestrado.pdf ▾ [Translate this page](#)

Sep 11, 2012 - candidatos ao Curso de **Mestrado** Profissional em Engenharia de ... (mesc@puro.uff.br), indicando nome completo e CPF do candidato.

[MESC - Mestrado em Educação e Sociedade do Conhecime...](#)

<https://woc.uc.pt/fpce/course/planocurricular.do?courseId=80> ▾

Ano, Unidade Curricular, Duração, Área, Ramos, ECTS. 1, Análise do Discurso Pedagógico Contemporâneo, 1º Sem, C. Educação, 6.0 ...

Contexto ... (2)

Problemas na Web Atual

- Pouca integração de informações
 - Site de loja realiza de venda de carros
 - Site de fabricante descreve dados dos carros
- Interação essencialmente homem-máquina
 - Obtenção “manual” de dados da web
- Redundância de dados
 - Representação de dados distinta
 - Possível inconsistência de dados
- Não classificação dos dados disponíveis ⁴

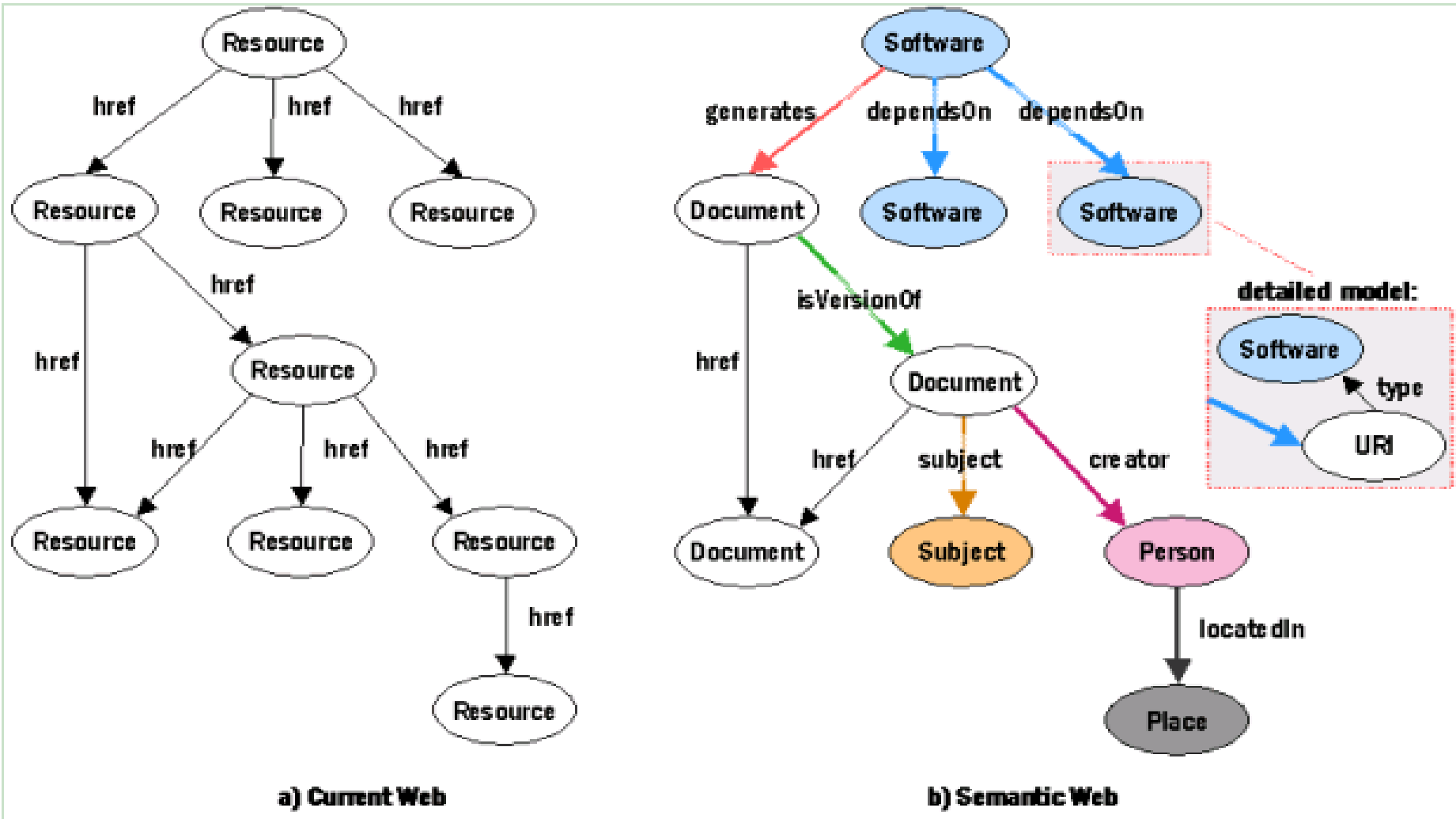
Algumas Iniciativas

- Disseminação de serviços
- Classificação e padronização dos dados
 - Definição de ontologias
 - Onto (o que existe) + Logos (conhecimento sobre)
- Uso de linguagens para representação dos dados
 - XML, RDF, RDFa, OWL, ...
- Disponibilização de repositórios (datasets) e vocabulários: DBPedia [4], GeoNames [3], DBLP, FOAF, ...

Algumas Iniciativas

- Schema.org
- WolframAlpha
 - Siri
- Google Knowledge Graph
- Open Graph Protocol (Facebook)
- Satori Knowledge Base (Bing, Microsoft)
- Yahoo!, Baidu, ...

Hoje e Amanhã [1]



O que é a Web Semântica

"The Semantic Web is an extension of the current web in which information is given well-defined meaning, better enabling computers and people to work in cooperation." [5]

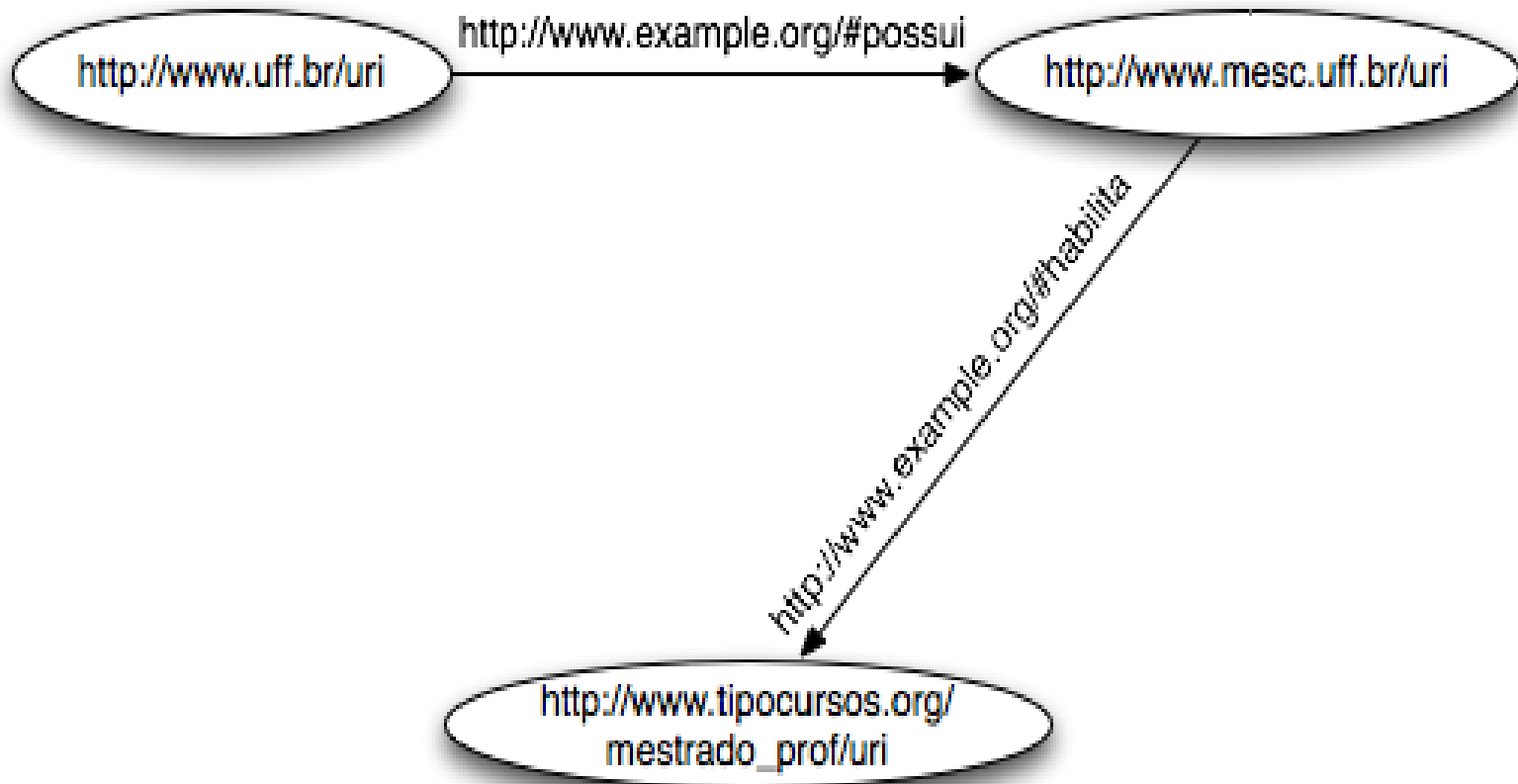
Problemas com XML

```
< cursos >  
  < nome > MESC < / nome >  
  < universidade > UFF < / universidade >  
< / cursos >
```

```
< universidade >  
  < nome > UFF < / nome >  
  < cursos >  
    < mestrado > MESC < / mestrado >  
  < / cursos >  
< / universidade >
```

RDF

- Idéia básica: uso de grafos direcionados como modelo de dados



RDF

- RDF (*Resource Description Framework*) é uma linguagem para expressão de informações de forma comum e processáveis por máquina
- É um modelo de dados:
 - Motivação inicial para a definição de metadados de páginas web
 - Provê informação estruturada
 - Sintaxe principal baseada em XML

RDF: Conceitos Básicos

- Bloco básico: tripla (objeto-atributo-valor)
 - É chamada de sentença (*statement*)
 - “*A UFF possui um curso chamado MESC*”
 - Objeto: UFF
 - Atributo: possui
 - Valor: Curso MESC

RDF:

Conceitos Básicos (2)

- Componentes fundamentais de RDF:
 - Recursos: qualquer coisa definida através de uma URI
 - `http://www.uff.br/uri`
 - Propriedades: recursos que descrevem uma relação (binária)
 - `universidades:possuiCurso`
 - Sentenças: associa um valor a uma propriedade de um recurso específico
 - `<#univ1102 universidades:possuiCurso "http://www.mesc.uff.br/uri">`

RDF/XML

- Um documento RDF pode ser representado por um elemento XML com a tag **rdf:RDF**
- O conteúdo desta tag é uma lista de descrições, as quais utilizam tags **rdf:Description**
- Cada descrição contém uma sentença para um recurso, identificado de 2 formas:
 - atributo **about**: ref. para uma descrição
 - atributo **ID**: criação de uma nova descrição¹⁴

```
<?xml version="1.0"?>
```

```
<rdf:RDF
```

```
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:cd="http://www.recshop.fake/cd#"
  xmlns:cntrs="http://www.planet.org/countries#">
```

```
  <rdf:Description rdf:about="http://www.recshop.fake/cd/Empire
```

```
  Burlesque">
```

```
    <cd:artist>Bob Dylan</cd:artist>
```

```
    <cntrs:country>USA</cntrs:country>
```

```
    <cd:company>Columbia</cd:company>
```

```
    <cd:price>10.90</cd:price>
```

```
    <cd:year>1985</cd:year>
```

```
  </rdf:Description>
```

```
  <rdf:Description rdf:about="http://www.recshop.fake/cd/Hide your heart">
```

```
    <cd:artist>Bonnie Tyler</cd:artist>
```

```
    <cntrs:country>UK</cntrs:country>
```

```
    <cd:company>CBS Records</cd:company>
```

```
    <cd:price>9.90</cd:price>
```

```
    <cd:year>1988</cd:year>
```

```
  </rdf:Description>
```

```
...
```

```
</rdf:RDF>
```

Exemplo em RDF [3]

N-triples

<http://www.recshop.fake/cd/Empire Burlesque">

<http://www.recshop.fake/cd/artist> "Bob Dylan"

<http://www.recshop.fake/cd/Empire Burlesque">

<http://www.planet.org/countries/country> "USA"

<http://www.recshop.fake/cd/Empire Burlesque">

<http://www.recshop.fake/cd/company> "Columbia"

<http://www.recshop.fake/cd/Empire Burlesque">

<http://www.recshop.fake/cd/price> "10.90"

<http://www.recshop.fake/cd/Empire Burlesque">

<http://www.recshop.fake/cd/year> "1985"

Exemplo em RDF [4]

Turtle

```
<http://www.recshop.fake/cd/Empire Burlesque">  
  <http://www.recshop.fake/cd/artist> "Bob Dylan" ;  
  <http://www.planet.org/countries/country> "USA" ;  
  <http://www.recshop.fake/cd/company> "Columbia" ;  
  <http://www.recshop.fake/cd/price> "10.90" ;  
  <http://www.recshop.fake/cd/year> "1985" .
```

SPARQL

- *Simple Protocol And RDF Query Language*
- Linguagem de consulta de documentos RDF
- Padronização similar a XQuery para XML

SPARQL – Exemplo

PREFIX

abc: <<http://mynamespace.com/example#>>

SELECT ?capital ?country

WHERE {

?x abc:cityname ?capital.

?y abc:countryname ?country.

?x abc:isCapitalOf ?y.

?y abc:isInContinent abc:africa.

}

SPARQL

- Variáveis são prefixadas com ?
- ?capital e ?country são os retornos
- O retorno da consulta são todos os dados que satisfazem as 4 triplas RDF (WHERE)

```
PREFIX
```

```
abc: <http://mynamespace.com/example#>
```

```
SELECT ?capital ?country
```

```
WHERE {
```

```
    ?x abc:cityname ?capital.
```

```
    ?y abc:countryname ?country.
```

```
    ?x abc:isCapitalOf ?y.
```

```
    ?y abc:isInContinent abc:africa.
```

```
}
```

SPARQL – Tutorial

<http://www.cambridgesemantics.com/semantic-university/sparql-by-example>

Linked Data

- Uma das principais aplicações de RDF
- Define boas práticas para publicação e conexão de dados estruturados na Web usando URIs e RDF
- Exemplos: DBpedia, GeoNames, US Census, EuroStat, MusicBrainz, BBC Programmes, Flickr, DBLP, PubMed, UniProt, FOAF, SIOC, OpenCyc, UMBEL, Virtual Observatories, freebase,...

GeoNames

GeoNames

www.geonames.org/maps/cities.html

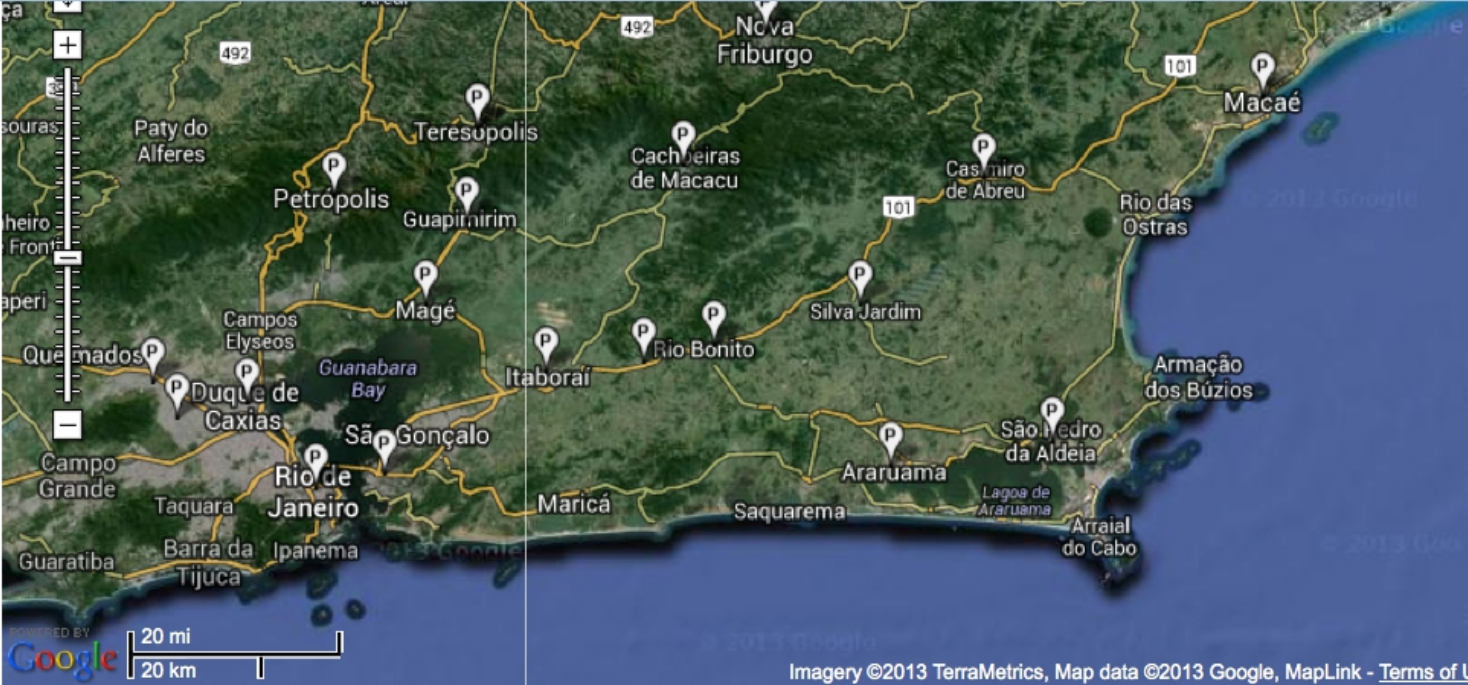
POWERED BY Google

Terms of Use

GeoNames Wikipedia

features

- ☒ city, village, ...
- ☐ mountain, hill, rock, ...
- ☐ stream, lake, ...
- ☐ country, state, region, ...
- ☐ parks, area, ...
- ☐ road, railroad
- ☐ spot, building, farm
- ☐ forest, heath, ...
- ☐ undersea



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only 10 objects displayed, zoom in or deselect some features

	Name	country	population	feature	km to center
1	Rio de Janeiro	Brazil	6023699	seat of a first-order administrative division	73.84 km
2	Nova Iguaçu	Brazil	1002118	seat of a second-order administrative division	93.51 km
3	Niterói	Brazil	456456	populated place	63.28 km
4	Duque de Caxias	Brazil	818329	populated place	79.97 km
5	Petrópolis	Brazil	272691	populated place	66.06 km

Localizar: free

☐ Diferenciar maiúsculas/minúsculas

GeoNames Web Services

GeoNames WebServices overview

	WebService	XML	JSON	RDF	CSV	TXT	RSS	KML
1	astergdem	XML	JSON			TXT		
2	children	XML	JSON					
3	cities	XML	JSON					
4	contains	XML	JSON					
5	countryCode	XML	JSON			TXT		
6	countryInfo	XML	JSON		CSV			
7	countrySubdivision	XML	JSON					
8	earthquakes	XML	JSON					
9	extendedFindNearby	XML						
10	findNearby	XML	JSON					
11	findNearbyPlaceName	XML	JSON					
12	findNearbyPostalCodes	XML	JSON					
13	findNearbyStreets 🇺🇸	XML	JSON					
14	findNearbyStreetsOSM	XML	JSON					
15	findNearByWeather	XML	JSON					
16	findNearbyWikipedia	XML	JSON				RSS	
17	findNearestAddress 🇺🇸	XML	JSON					
18	findNearestIntersection 🇺🇸	XML	JSON					
19	findNearestIntersectionOSM	XML	JSON					

GeoNames

Web Services

Places

Cities and Placenames

Webservice Type : REST

Url : api.geonames.org/citiesJSON?

Parameters :

north,south,east,west : coordinates of bounding box

callback : name of javascript function (optional parameter)

lang : language of placenames and wikipedia urls (default = en)

maxRows : maximal number of rows returned (default = 10)

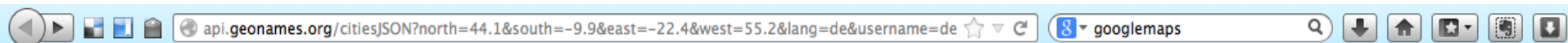
Result : returns a list of cities and placenames in the bounding box, ordered by relevancy (capital/population). Placenames close together are filterered out and only the larger name is included in the resulting list.

Example : <http://api.geonames.org/citiesJSON?north=44.1&south=-9.9&east=-22.4&west=55.2&lang=de&username=demo>

This service is also available in XML output :

Example : <http://api.geonames.org/cities?north=44.1&south=-9.9&east=-22.4&west=55.2&username=demo>

GeoNames Web Services



```
{
  "geonames": [
    {
      "fcodeName": "capital of a political entity",
      "toponymName": "Mexico City",
      "countrycode": "MX",
      "fcl": "P",
      "fclName": "city, village, ...",
      "name": "Mexiko-Stadt",
      "wikipedia": "en.wikipedia.org/wiki/Mexico_City",
      "lng": -99.12766456604,
      "fcode": "PPLC",
      "geonameId": 3530597,
      "lat": 19.428472427036,
      "population": 12294193
    },
    {
      "fcodeName": "capital of a political entity",
      "toponymName": "Manila",
      "countrycode": "PH",
      "fcl": "P",
      "fclName": "city, village, ...",
      "name": "Manila",
      "wikipedia": "en.wikipedia.org/wiki/Manila",
      "lng": 120.9822,
      "fcode": "PPLC",
      "geonameId": 1701668,
      "lat": 14.6042,
      "population": 10444527
    },
    {
      "fcodeName": "capital of a political entity",
      "toponymName": "Dhaka",
      "countrycode": "BD",
      "fcl": "P",
      "fclName": "city, village, ...",
      "name": "Dhaka",
      "wikipedia": "en.wikipedia.org/wiki/Dhaka",
      "lng": 90.40743827819824,
      "fcode": "PPLC",
      "geonameId": 1185241,
      "lat": 23.710395616597037,
      "population": 10356500
    },
    {
      "fcodeName": "capital of a political entity",
      "toponymName": "Seoul",
      "countrycode": "KR",
      "fcl": "P",
      "fclName": "city, village, ...",
      "name": "Seoul",
      "wikipedia": "en.wikipedia.org/wiki/Seoul",
      "lng": 126.977834701538,
      "fcode": "PPLC",
      "geonameId": 1835848,
      "lat": 37.5682561388953,
      "population": 10349312
    },
    {
      "fcodeName": "capital of a political entity",
      "toponymName": "Jakarta",
      "countrycode": "ID",
      "fcl": "P",
      "fclName": "city, village, ...",
      "name": "Jakarta",
      "wikipedia": "en.wikipedia.org/wiki/Jakarta",
      "lng": 106.84513092041016,
      "fcode": "PPLC",
      "geonameId": 1642911,
      "lat": -6.214623197035775,
      "population": 8540121
    },
    {
      "fcodeName": "capital of a political entity",
      "toponymName": "Tokyo",
      "countrycode": "JP",
      "fcl": "P",
      "fclName": "city, village, ...",
      "name": "Tokyo",
      "wikipedia": "en.wikipedia.org/wiki/Tokyo",
      "lng": 139.69171,
      "fcode": "PPLC",
      "geonameId": 1850147,
      "lat": 35.6895,
      "population": 8336599
    },
    {
      "fcodeName": "capital of a political entity",
      "toponymName": "Taipei",
      "countrycode": "TW",
      "fcl": "P",
      "fclName": "city, village, ...",
      "name": "Taipeh",
      "wikipedia": "en.wikipedia.org/wiki/Taipei_Railway_Station",
      "lng": 121.531846,
      "fcode": "PPLC",
      "geonameId": 1668341,
      "lat": 25.047763,
      "population": 7871900
    },
    {
      "fcodeName": "capital of a political entity",
      "toponymName": "Bogotá",
      "countrycode": "CO",
      "fcl": "P",
      "fclName": "city, village, ...",
      "name": "Bogotá",
      "wikipedia": "en.wikipedia.org/wiki/Bogotá",
      "lng": -74.08175468444824,
      "fcode": "PPLC",
      "geonameId": 3688689,
      "lat": 4.609705849789108,
      "population": 7674366
    },
    {
      "fcodeName": "capital of a political entity",
      "toponymName": "Beijing",
      "countrycode": "CN",
      "fcl": "P",
      "fclName": "city, village, ...",
      "name": "Peking",
      "wikipedia": "en.wikipedia.org/wiki/Beijing",
      "lng": 116.397228240967,
      "fcode": "PPLC",
      "geonameId": 1816670,
      "lat": 39.9074977414405,
      "population": 7480601
    },
    {
      "fcodeName": "capital of a political entity",
      "toponymName": "Hong Kong",
      "countrycode": "HK",
      "fcl": "P",
      "fclName": "city, village, ...",
      "name": "Hong Kong",
      "wikipedia": "en.wikipedia.org/wiki/Hong_Kong",
      "lng": 114.157691001892,
      "fcode": "PPLC",
      "geonameId": 1819729,
      "lat": 22.2855225817732,
      "population": 7012738
    }
  ]
}
```

GeoSPARQL

PREFIX co: <<http://www.geonames.org/countries/#>>

PREFIX xsd: <<http://www.w3.org/2001/XMLSchema#>>

PREFIX geo: <http://www.w3.org/2003/01/geo/wgs84_pos#>

SELECT ?link ?name ?pop ?lat ?lon

WHERE {

 ?link gs:within(-23.024132 -43.690338 -21.591043 -41.089325) .

 ?link gn:name ?name .

 ?link gn:population ?pop .

 ?link geo:lat ?lat .

 ?link geo:long ?lon

}

<http://geosparql.org/>

GeoSPARQL

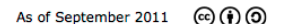
Resultado

Query=4332002 | [XML](#) | [KML](#) | [Home](#)

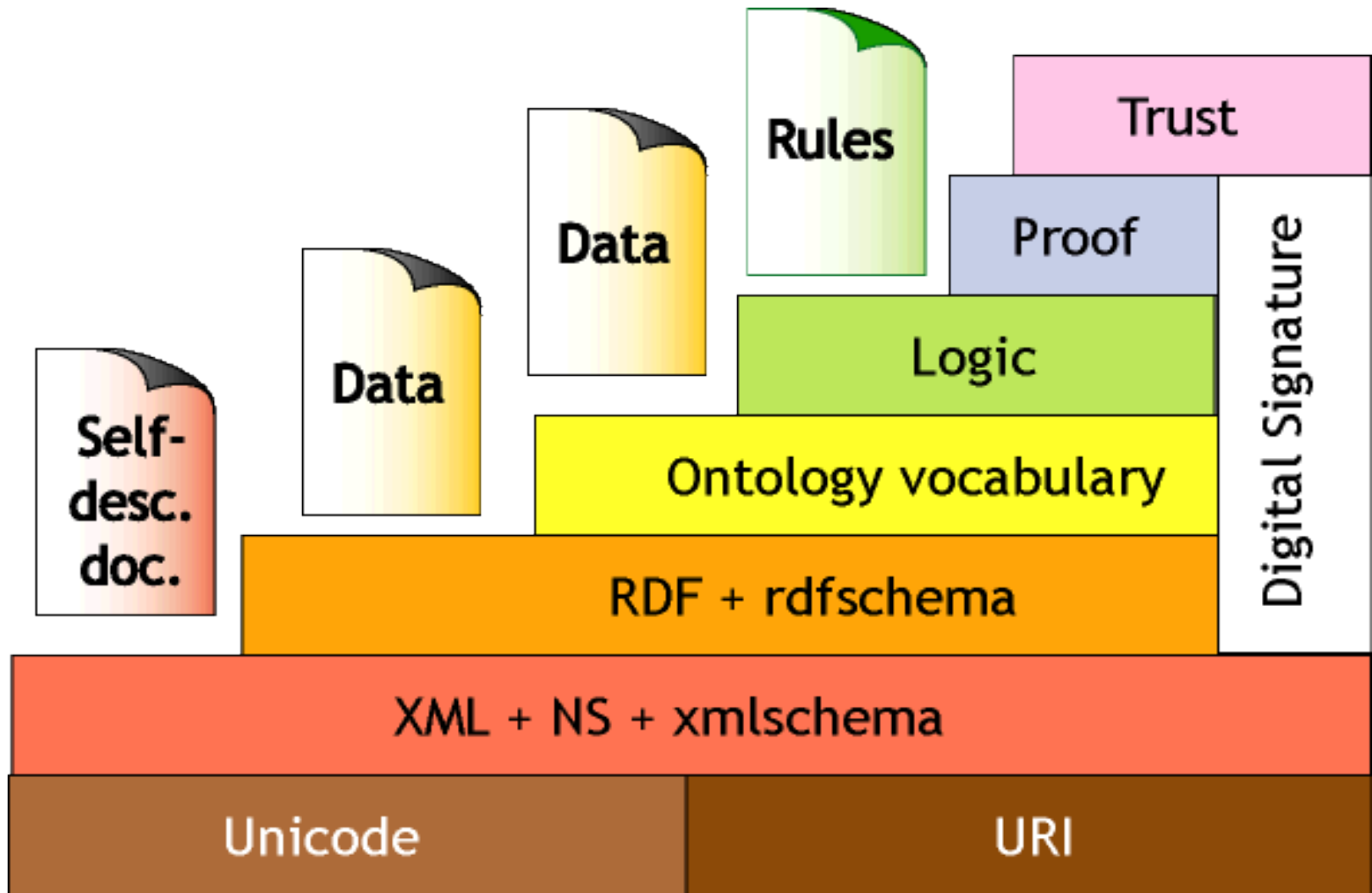
link	name	pop	lat	lon
<http://sws.geonames.org/3445433/>	"Vassouras"	"21174"	"-22.40388889"	"-43.6625"
<http://sws.geonames.org/3460132/>	"Japeri"	"95101"	"-22.64305556"	"-43.65333333"
<http://sws.geonames.org/3452073/>	"Queimados"	"135741"	"-22.71611111"	"-43.55527778"
<http://sws.geonames.org/3457191/>	"Miguel Pereira"	"23850"	"-22.45388889"	"-43.46888889"
<http://sws.geonames.org/3456160/>	"Nova Iguaçu"	"1002118"	"-22.75916667"	"-43.45111111"
<http://sws.geonames.org/3454827/>	"Paty do Alferes"	"20659"	"-22.42861111"	"-43.41861111"
<http://sws.geonames.org/3456290/>	"Nilópolis"	"147281"	"-22.8075"	"-43.41388889"
<http://sws.geonames.org/3448877/>	"São João de Meriti"	"454849"	"-22.80388889"	"-43.37222222"
<http://sws.geonames.org/3459505/>	"Juiz de Fora"	"470193"	"-21.76416667"	"-43.35027778"
<http://sws.geonames.org/3464374/>	"Duque de Caxias"	"818329"	"-22.78555556"	"-43.31166667"
<http://sws.geonames.org/3455141/>	"Paraíba do Sul"	"35517"	"-22.16194444"	"-43.29277778"
<http://sws.geonames.org/3446065/>	"Três Rios"	"71944"	"-22.11666667"	"-43.20916667"
<http://sws.geonames.org/3451190/>	"Rio de Janeiro"	"6023699"	"-22.90277778"	"-43.2075"
<http://sws.geonames.org/3454031/>	"Petrópolis"	"272691"	"-22.505"	"-43.17861111"
<http://sws.geonames.org/3456283/>	"Niterói"	"456456"	"-22.88333333"	"-43.10361111"
<http://sws.geonames.org/3461949/>	"Guapimirim"	"31202"	"-22.53722222"	"-42.98194444"
<http://sws.geonames.org/3446606/>	"Teresópolis"	"123979"	"-22.41222222"	"-42.96555556"
<http://sws.geonames.org/3460950/>	"Itaboraí"	"182498"	"-22.74444444"	"-42.85944444"
<http://sws.geonames.org/3457708/>	"Maricá"	"79551"	"-22.91944444"	"-42.81861111"
<http://sws.geonames.org/3446974/>	"Tanguá"	"23740"	"-22.73027778"	"-42.71416667"
<http://sws.geonames.org/3451261/>	"Rio Bonito"	"35997"	"-22.70861111"	"-42.60972222"
<http://sws.geonames.org/3456166/>	"Nova Friburgo"	"153361"	"-22.28194444"	"-42.53111111"
<http://sws.geonames.org/3448011/>	"Saquarema"	"62056"	"-22.92"	"-42.51027778"
<http://sws.geonames.org/3447591/>	"Silva Jardim"	"16888"	"-22.65083333"	"-42.39166667"
<http://sws.geonames.org/3465527/>	"Cordeiro"	"15601"	"-22.02861111"	"-42.36083333"
<http://sws.geonames.org/3466763/>	"Casimiro de Abreu"	"19087"	"-22.48055556"	"-42.20416667"
<http://sws.geonames.org/3448351/>	"São Pedro da Aldeia"	"55014"	"-22.83916667"	"-42.10277778"
<http://sws.geonames.org/3460774/>	"Itaocara"	"16762"	"-21.66916667"	"-42.07611111"
<http://sws.geonames.org/3451205/>	"Rio das Ostras"	"46618"	"-22.52694444"	"-41.945"
<http://sws.geonames.org/3458266/>	"Macaé"	"143029"	"-22.37083333"	"-41.78694444"
<http://sws.geonames.org/3449195/>	"São Fidélis"	"27793"	"-21.64611111"	"-41.74694444"
<http://sws.geonames.org/3467693/>	"Campos"	"387417"	"-21.75"	"-41.3"
<http://sws.geonames.org/3468425/>	"Cachoeiras de Macacu"	"46177"	"-22.4625"	"-42.65305556"
<http://sws.geonames.org/3468615/>	"Cabo Frio"	"108239"	"-22.87944444"	"-42.01861111"
<http://sws.geonames.org/3470142/>	"Belford Roxo"	"466096"	"-22.76416667"	"-43.39944444"
<http://sws.geonames.org/3471451/>	"Arraial do Cabo"	"26163"	"-22.96611111"	"-42.02777778"
<http://sws.geonames.org/3471487/>	"Armação de Búzios"	"23463"	"-22.74694444"	"-41.88166667"
<http://sws.geonames.org/3471715/>	"Araruama"	"109637"	"-22.87277778"	"-42.34305556"
<http://sws.geonames.org/3472609/>	"Além Paraíba"	"33907"	"-21.88777778"	"-42.70444444"

Linked Data x Web API

- Muitas fontes de dados da web, como Amazon, Ebay, Twitter, Google, oferecem acesso a seus dados através de APIs
- Estes dados são informados por inúmeras maneiras
- Linked Data utiliza um conjunto restrito de tecnologias para publicação de dados:
 - URIs para identificação
 - HTTP como mecanismo de acesso



Camadas da Web Semântica [1]



Aplicações

www.bbc.co.uk/music/artists/3a5d1cc7-627e-48ea-aba3-fdd20d782c33

dbpedia mobile

BBC Sign in News Sport Weather iPlayer TV Radio More... Search

iPlayer Radio WHAT'S NEW? Stations Categories Programmes + Favourites

MUSIC
HOME | SHOWCASE | GENRES

Chico Buarque
Born 19 junho 1944.

PLAYED MOST ON **BBC RADIO 3**

Share This Page
Share f t

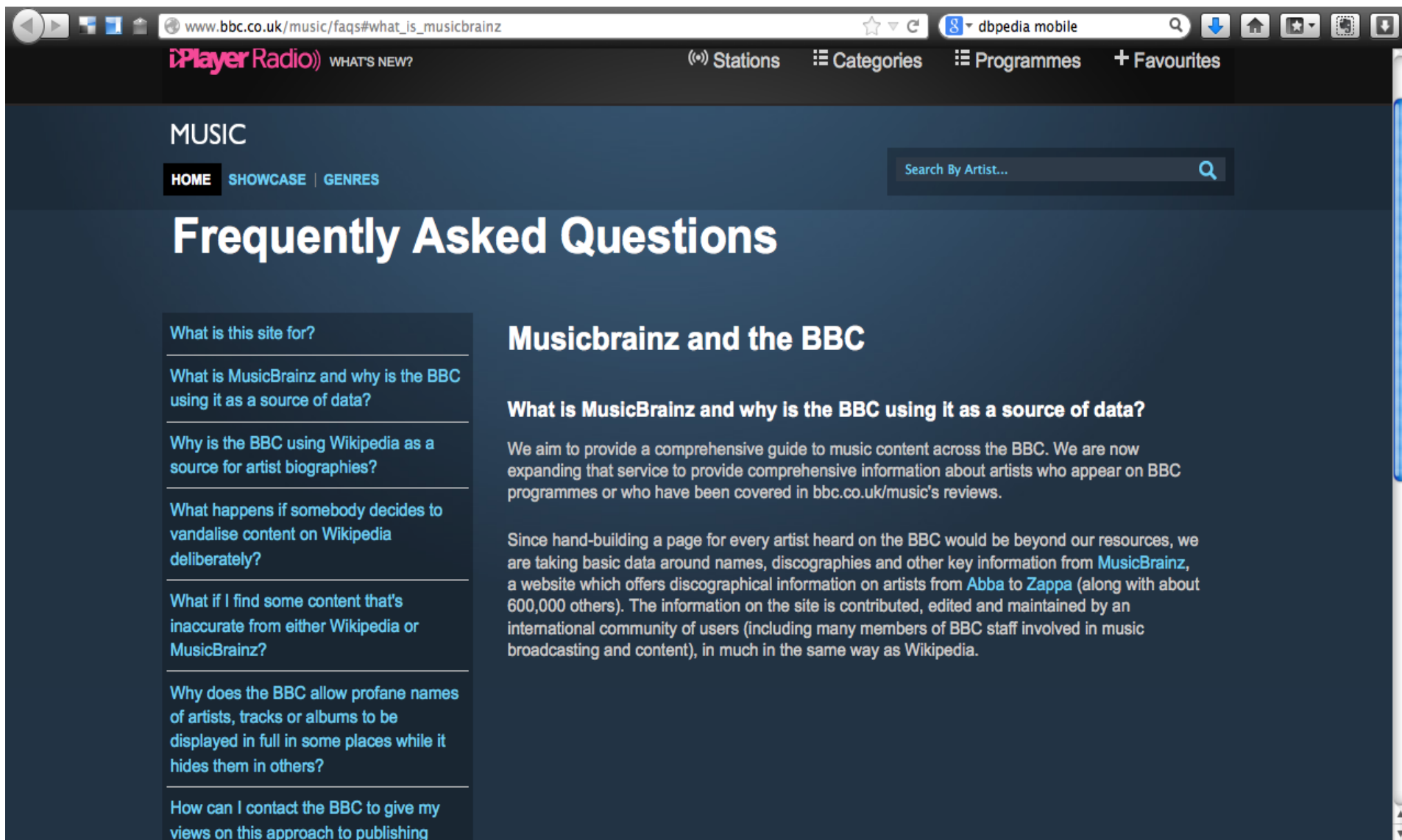
More BBC Music Highlights

Latest Tracks Played On The BBC

Biography



Aplicações



The screenshot shows a web browser window with the address bar displaying www.bbc.co.uk/music/faqs#what_is_musicbrainz. The page header includes the iPlayer Radio logo and navigation links for Stations, Categories, Programmes, and Favourites. The main content area is titled "Frequently Asked Questions" and features a sidebar with a list of questions and a main text area with answers.

MUSIC

HOME | SHOWCASE | GENRES

Search By Artist...

Frequently Asked Questions

What is this site for?

What is MusicBrainz and why is the BBC using it as a source of data?

Why is the BBC using Wikipedia as a source for artist biographies?

What happens if somebody decides to vandalise content on Wikipedia deliberately?

What if I find some content that's inaccurate from either Wikipedia or MusicBrainz?

Why does the BBC allow profane names of artists, tracks or albums to be displayed in full in some places while it hides them in others?

How can I contact the BBC to give my views on this approach to publishing

Musicbrainz and the BBC

What is MusicBrainz and why is the BBC using it as a source of data?

We aim to provide a comprehensive guide to music content across the BBC. We are now expanding that service to provide comprehensive information about artists who appear on BBC programmes or who have been covered in bbc.co.uk/music's reviews.

Since hand-building a page for every artist heard on the BBC would be beyond our resources, we are taking basic data around names, discographies and other key information from MusicBrainz, a website which offers discographical information on artists from Abba to Zappa (along with about 600,000 others). The information on the site is contributed, edited and maintained by an international community of users (including many members of BBC staff involved in music broadcasting and content), in much in the same way as Wikipedia.

Aplicações

musicbrainz.org/artist/3a5d1cc7-627e-48ea-aba3-fdd20d782c33/details

search Artist Search

About Blog Products Search Documentation Contact Us Log In Create Account

 **Chico Buarque**
~ Person

Overview Releases Recordings Works Relationships Aliases Tags Details Edit

Details

Name: Chico Buarque
MBID: 3a5d1cc7-627e-48ea-aba3-fdd20d782c33
Last updated: 2013-07-24 07:00 UTC
Permanent link: <http://musicbrainz.org/artist/3a5d1cc7-627e-48ea-aba3-fdd20d782c33>
XML: <http://musicbrainz.org/ws/2/artist/3a5d1cc7-627e-48ea-aba3-fdd20d782c33?inc=aliases>

Artist information
Sort name: Buarque, Chico
Type: Person
Gender: Male
Born: 1944-06-19 (69 years ago)
Area: Brazil
IPI code: 00040160436
IPI code: 00064390379

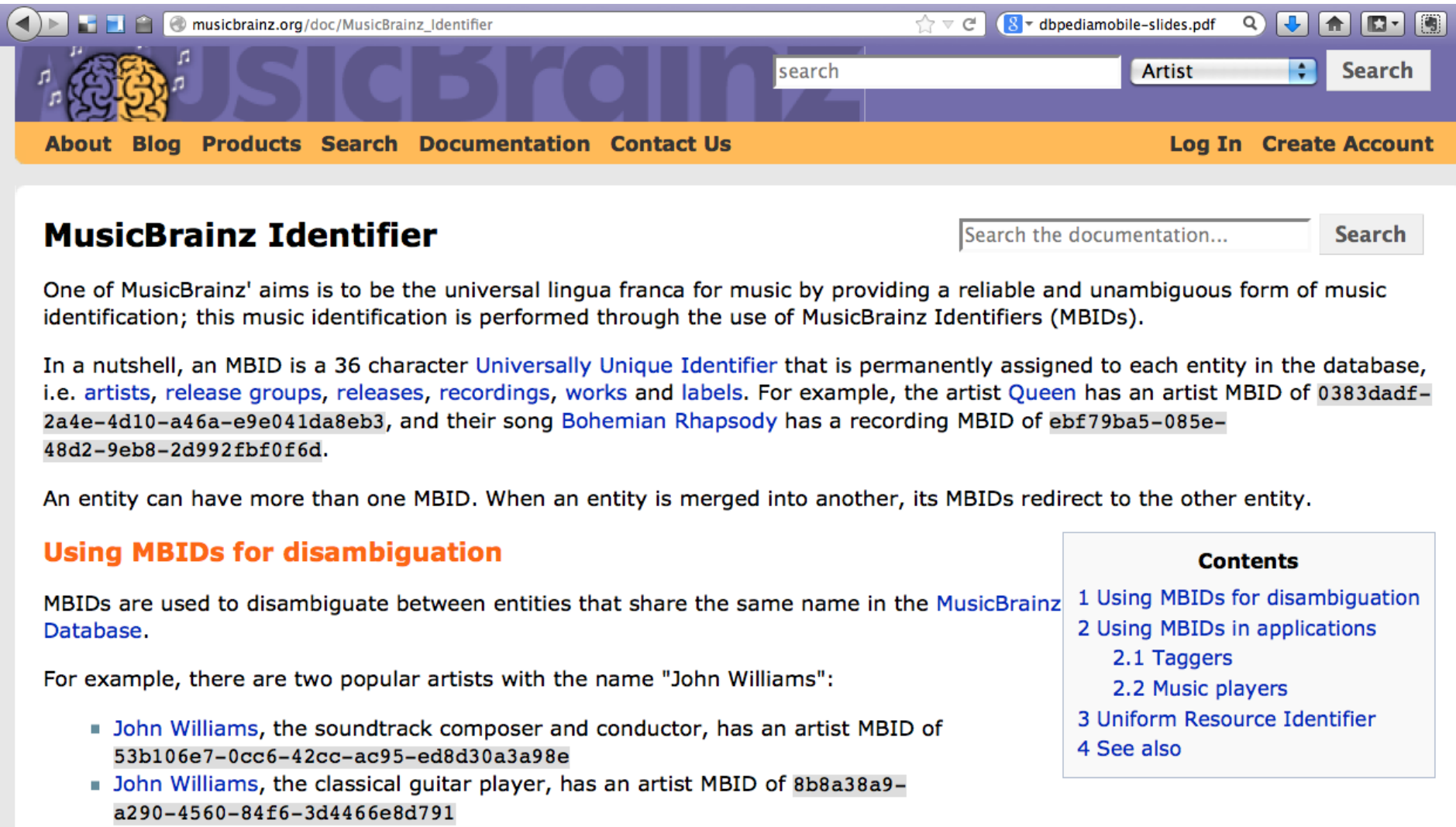
Rating
★★★★★ (see all ratings)

Tags
mpb, brazilian, playwright, singer, singer/songwriter, more...

Editing
■ [Log in to edit](#)

External links

Aplicações



The screenshot shows the MusicBrainz website interface. At the top, there's a navigation bar with links: About, Blog, Products, Search, Documentation, Contact Us. On the right, there are links for Log In and Create Account. Below the navigation bar, the main heading is "MusicBrainz Identifier". To the right of this heading is a search box labeled "Search the documentation..." with a "Search" button. The main content area contains a paragraph explaining the purpose of MusicBrainz and MBIDs. It states that MusicBrainz aims to be a universal lingua franca for music by providing reliable and unambiguous identification through MBIDs. It then defines an MBID as a 36-character Universally Unique Identifier assigned to each entity in the database, with examples for the artist Queen and the song Bohemian Rhapsody. A second paragraph notes that an entity can have more than one MBID and that MBIDs redirect when entities are merged. A section titled "Using MBIDs for disambiguation" follows, explaining that MBIDs are used to distinguish between entities with the same name. It provides two examples of artists named "John Williams" with their respective MBIDs. On the right side of the page, there is a "Contents" box listing the following items: 1 Using MBIDs for disambiguation, 2 Using MBIDs in applications (with sub-items 2.1 Taggers and 2.2 Music players), 3 Uniform Resource Identifier, and 4 See also.

musicbrainz.org/doc/MusicBrainz_Identifier

search Artist Search

About Blog Products Search Documentation Contact Us Log In Create Account

MusicBrainz Identifier

Search the documentation... Search

One of MusicBrainz' aims is to be the universal lingua franca for music by providing a reliable and unambiguous form of music identification; this music identification is performed through the use of MusicBrainz Identifiers (MBIDs).

In a nutshell, an MBID is a 36 character [Universally Unique Identifier](#) that is permanently assigned to each entity in the database, i.e. [artists](#), [release groups](#), [releases](#), [recordings](#), [works](#) and [labels](#). For example, the artist [Queen](#) has an artist MBID of `0383dadf-2a4e-4d10-a46a-e9e041da8eb3`, and their song [Bohemian Rhapsody](#) has a recording MBID of `ebf79ba5-085e-48d2-9eb8-2d992fbf0f6d`.

An entity can have more than one MBID. When an entity is merged into another, its MBIDs redirect to the other entity.

Using MBIDs for disambiguation

MBIDs are used to disambiguate between entities that share the same name in the [MusicBrainz Database](#).

For example, there are two popular artists with the name "John Williams":

- [John Williams](#), the soundtrack composer and conductor, has an artist MBID of `53b106e7-0cc6-42cc-ac95-ed8d30a3a98e`
- [John Williams](#), the classical guitar player, has an artist MBID of `8b8a38a9-a290-4560-84f6-3d4466e8d791`

Contents

- 1 Using MBIDs for disambiguation
- 2 Using MBIDs in applications
 - 2.1 Taggers
 - 2.2 Music players
- 3 Uniform Resource Identifier
- 4 See also

Aplicações

DBPedia Mobile [8]



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